

# John Ryan

+447340934016 • [john.patrick.ryan.ie@gmail.com](mailto:john.patrick.ryan.ie@gmail.com) • [github.com/johnryan465](https://github.com/johnryan465) • [linkedin.com/in/johnryanie](https://linkedin.com/in/johnryanie)

## SUMMARY

Senior Research Engineer with more than 8 years of machine learning experience, beginning with the development of production NLP systems in 2016. Combines a track record of founding deep-tech ventures with recent leadership in large-scale AI for drug discovery at Isomorphic Labs. Proven ability to lead technical projects from initial concept through to production.

## EXPERIENCE

### ISOMORPHIC LABS | SENIOR RESEARCH ENGINEER - PERFORMANCE AND SCALING

Jan 2026 - Present | London, UK

- **Executed core research** on frontier AI models, alongside developing advanced performance and scaling solutions to push the boundaries of computational efficiency.
- **Enabled** compatibility for core models on a new class of hardware, unlocking large-scale inference on thousands of accelerators and enhancing computational flexibility.
- **Scaled** single training jobs to run efficiently on high-hundreds of accelerators by resolving complex multi-node communication bottlenecks/issues.

### | RESEARCH ENGINEER II - APPLIED ML RESEARCH

Nov 2024 - Dec 2025

- **Directed** the technical strategy and roadmap for the inference domain within the Applied Research team, establishing best practices and guiding the direction of key engineering efforts.
- **Architected and spearheaded** the development of a scalable internal platform for repeatable ML inference workflows for repeatable science, now used by 4+ teams.
- **Engineered** a deep optimization initiative for core inference pipelines, reducing end-to-end latency by over 98% (from >10 minutes to <10 seconds) through custom kernel development and targeted profiling.
- **Co-authored** the technical report detailing the proprietary IsoDDE for end-to-end drug discovery, achieving state-of-the-art performance in protein-ligand structure prediction and binding affinity.

### | RESEARCH ENGINEER - APPLIED ML RESEARCH

Feb 2024 - Oct 2024

- **Led** machine learning for two drug discovery programs, establishing a robust feedback loop between AI prediction and experimental validation to prioritize high-potential therapeutic candidates.
- **Onboarded and mentored** several new engineers, providing technical guidance to accelerate their contributions to core projects and their transition to drug discovery.
- **Performed and implemented research** on various topics, including experimental conditioning, evaluation protocols for novel tasks, inference time guidance and improving model computational performance.

### ENTREPRENEUR FIRST | FOUNDER IN RESIDENCE

Oct 2023 - Jan 2024 | London, UK

- Selected for the globally competitive talent investor program to conceptualize and build a deep-tech venture from the ground up.
- Drove rapid ideation and prototyping of novel AI applications, assessing technical viability and architecting alternative solutions to solve complex problems.

### DUCTILE AI | FOUNDER

July 2022 - Sep 2023 | London, UK

- **Engineered** robust Visual SLAM systems for resource-constrained aerial drones; rapidly advanced from research to functional prototypes deployed for field testing in Ukraine and Estonia.
- **Drove** the research strategy, evaluating and implementing novel approaches combining Neural Radiance Fields (NeRFs) and Bayesian Modelling.
- **Architected** and optimized the full MLOps pipeline using ROS, demonstrating deep technical leadership in a production-focused environment.

### RIGR AI | RESEARCH LEAD

June 2022 - Aug 2023 | London, UK (Contract)

- **Directed** multiple concurrent R&D projects in NLP, Computer Vision, and Reinforcement Learning for external clients.

- **Owned** the deployment of research models into production, focusing on scalability and performance optimization.
- **Served** as a key technical partner in the €2M EU-funded ARICA consortium, developing the Video Summarisation Tool (VST) to help law enforcement investigate and prioritize high-risk cases on the darknet.

## LINKEDIN | MACHINE LEARNING ENGINEERING INTERN

Summers 2019, 2020, 2021 | Dublin, Ireland

- **Developed and productionized** active learning and few-shot learning systems for NLP using Bayesian ML, PyTorch, and internal distributed training frameworks.
- **Architected** a scalable ML pipeline for large-scale personalized generative text models using Spark, Tensorflow, and BERT.
- **Built** and delivered the core tooling for machine learning interpretability to debug models and datasets, integrated with Tensorflow and Tensorboard.

## TEAMWORK.COM | SOFTWARE DEVELOPER

Summers 2017 & 2018 | Cork, Ireland

- Engineered and shipped new backend features for the Teamwork Desk platform, and explored early-stage applications of NLP to enhance product functionality. (2018)
- Built and benchmarked a proof-of-concept text similarity search engine using Lucene and text embeddings to evaluate its potential for new product features. (2017)

## INTEL SECURITY | DATA SCIENCE INTERN

Summer 2016 | Cork, Ireland

- Developed production NLP systems using Gradient Boosted Trees & CNNs; this work directly led to co-authoring and presenting an academic paper on online harassment detection in 2017.

## PUBLICATIONS & RESEARCH

### ISOMORPHIC LABS DRUG DESIGN ENGINE (ISODDE) | TECHNICAL REPORT | 2026

- Co-authored the technical report detailing the proprietary IsoDDE for end-to-end drug discovery, achieving state-of-the-art performance in protein-ligand structure prediction and binding affinity.

### REINFORCEMENT LEARNING FOR SPACE OPERATIONS | MSc THESIS | 2022

- Designed and implemented a fully differentiable simulator for satellite operations on GPUs to enable gradient-based learning.
- Developed and benchmarked safe reinforcement learning policies for satellite control in collaboration with the European Space Agency (ESA).

### BATCH ACTIVE LEARNING WITH DEEP KERNEL GAUSSIAN PROCESS CLASSIFIERS | UNDERGRADUATE THESIS (DISTINCTION) | 2021

- Implemented and benchmarked Bayesian Active Learning methods with Deep Gaussian Process models in PyTorch.
- Devised and proved a method to improve the computational complexity of batch selection, making it viable for larger datasets.

### HACK HARASSMENT: TECHNOLOGY SOLUTIONS TO COMBAT ONLINE HARASSMENT | ALW1: 1ST WORKSHOP ON ABUSIVE LANGUAGE ONLINE | 2017

- Co-authored and presented work on production NLP systems using Gradient Boosted Trees and CNNs for harassment detection.

## EDUCATION

### UNIVERSITY OF OXFORD | MASTERS - COMPUTER SCIENCE - MERIT

Oct 2021 - June 2022

BACHELORS - COMPUTER SCIENCE - FIRST CLASS HONOURS

Oct 2018- June 2021

## SKILLS

**Languages:** Python, C++, CUDA, Golang, Javascript, Julia, Java, Scala, Bash

**AI Frameworks & Libraries:** PyTorch, JAX, Tensorflow, Triton, XLA, Pallas, Keras

**Distributed Systems & MLOps:** Docker, Kubernetes, DDP, FSDP, ROS, Linux, Spark

## HONORS & AWARDS

### SELECTED AWARDS

- Winner - OxfordHack @ Major League Hacking (2018)
- 1st Place Overall & 1st Place Junior - Irish Collegiate Programming Competition (IrICPC) (2018)
- Irish National Delegate - International Olympiad in Informatics (IOI), Japan (2018) & Iran (2017) & Russia (2016)